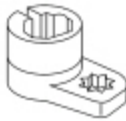


Engine Control Components — Exploded View, 5.0L (4V)

 [Ford Ecat Electronic Parts Catalog](#)

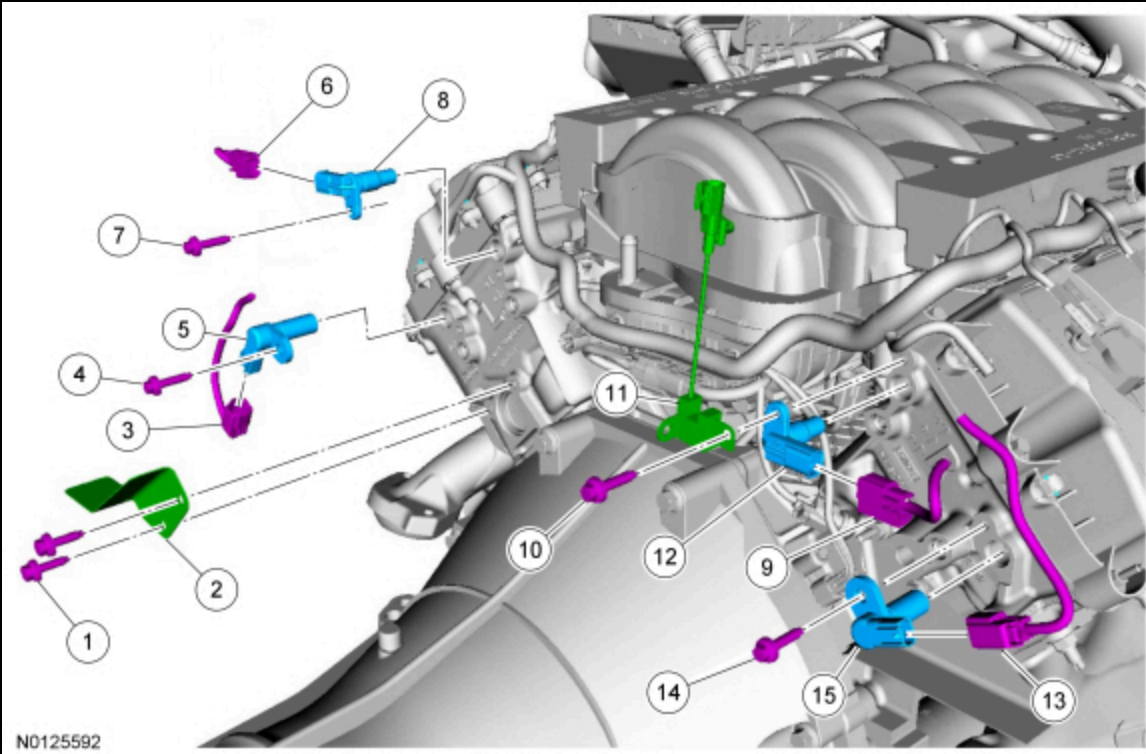
Special Tool(s)

 ST1447-A	Socket, Exhaust Gas Oxygen Sensor 303-476 (T94P-9472-A)
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Material

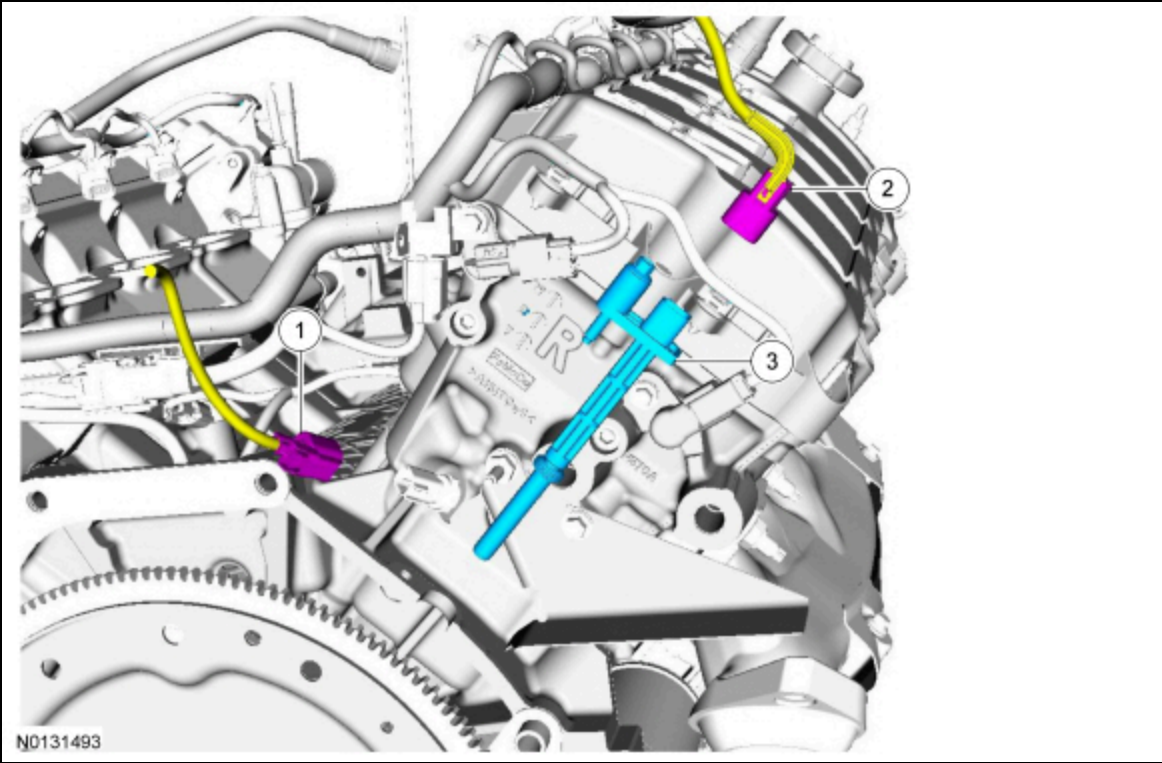
Item	Specification
Motorcraft® High Temperature Nickel Anti-Seize Lubricant XL-2	—
Motorcraft® Penetrating and Lock Lubricant (US); Motorcraft® Penetrating Fluid (Canada) XL-1 (US); CXC-51-A (Canada)	—

Camshaft Position (CMP) Sensor



Item	Part Number	Description
1	—	LH heat shield bolt - 10 Nm (89 lb-in) (2 required)
2	39686 39686	LH heat shield
3	—	LH exhaust Camshaft Position (CMP) sensor electrical connector (part of 12A581)
4	W503278 W503278	LH exhaust CMP sensor bolt - 10 Nm (89 lb-in)
5	6B2886 B288	LH exhaust CMP sensor
6	—	LH intake CMP sensor electrical connector (part of 12A581)
7	W503278 W503278	LH intake CMP sensor bolt - 10 Nm (89 lb-in)
8	12K073 12K073	LH intake CMP sensor
9	—	RH intake CMP sensor electrical connector (part of 12A581)
10	W503278 W503278	RH intake CMP sensor bolt - 10 Nm (89 lb-in)
11	18801 18801	RH radio ignition interference capacitor
12	12K073 12K073	RH intake CMP sensor
13	—	RH exhaust CMP sensor electrical connector (part of 12A581)
14	W503278 W503278	RH exhaust CMP sensor bolt - 10 Nm (89 lb-in)
15	6B2886 B288	RH exhaust CMP sensor

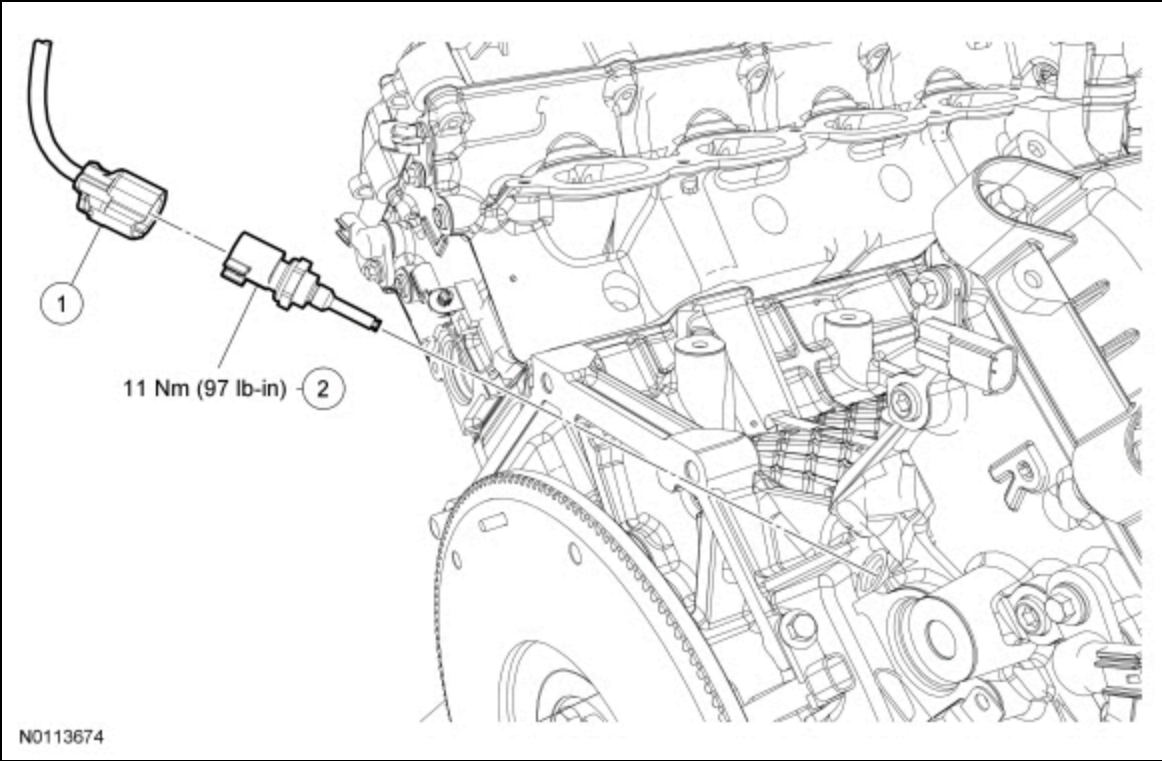
Crankshaft Position (CKP) Sensor



Item	Part Number	Description
1	—	Cylinder Head Temperature (CHT) sensor electrical connector (part of 12A581)
2	—	Crankshaft Position (CKP) sensor electrical connector (part of 12A581)
3	6C3156C315	CKP sensor and bolt assembly - 10 Nm (89 lb-in)

Cylinder Head Temperature (CHT) Sensor

NOTE: It is necessary to remove the intake manifold to perform this procedure. For additional information, refer to Section 303-01C.



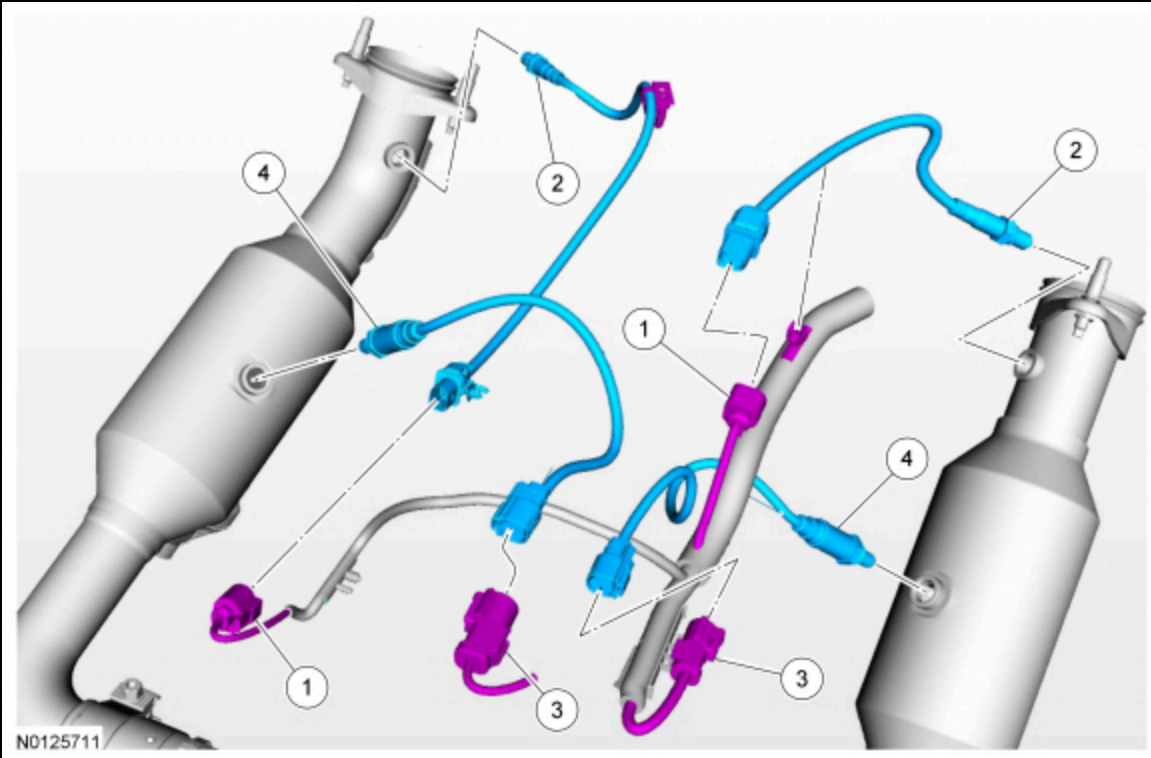
Item	Part Number	Description
1	—	Cylinder Head Temperature (CHT) sensor electrical connector (part of 12A581)
2	6G0046G004	CHT sensor

Heated Oxygen Sensor (HO2S) and Catalyst Monitor Sensor (CMS)

NOTE: If necessary, lubricate the Heated Oxygen Sensor (HO2S) and/or Catalyst Monitor Sensor (CMS) with penetrating and lock lubricant to assist in removal.

NOTE: Apply anti-seize to the threads of the HO2S and/or CMS prior to installing.

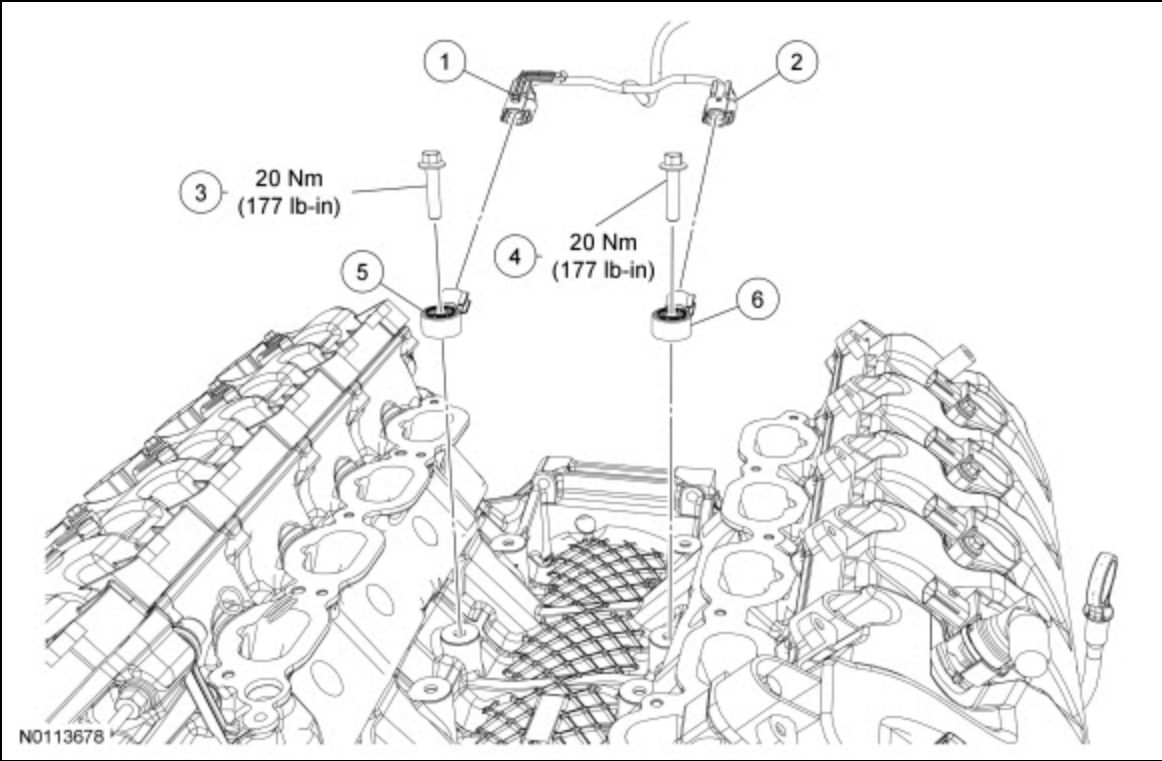
NOTE: The correct torque wrench setting must be calculated when using the Exhaust Gas Oxygen Sensor Socket to install the HO2S and/or CMS. [Refer to the Torque Wrench Adapter Formulas in the Appendix.](#)



Item	Part Number	Description
1	—	Heated Oxygen Sensor (HO2S) electrical connectors (part of 15525)
2	9F4729F472	HO2S
3	—	Catalyst Monitor Sensor (CMS) electrical connectors (part of 15525)
4	9G4449G444	CMS

Knock Sensor (KS)

NOTE: It is necessary to remove the intake manifold to perform this procedure. For additional information, refer to Section 303-01C.



Item	Part Number	Description
1	—	RH Knock Sensor (KS) electrical connector (part of 14B485)
2	—	LH KS electrical connector (part of 14B485)
3	W500110 W500110	RH KS bolt
4	W500110 W500110	LH KS bolt
5	12A699 12A699	RH KS
6	12A699 12A699	LH KS

Variable Camshaft Timing (VCT) Variable Force Solenoid

NOTICE: The Variable Camshaft Timing (VCT) variable force solenoid pins must be fully depressed to avoid interference with the VCT valve tips when installing the solenoids. Failure to follow these instructions can result in damage to the engine.

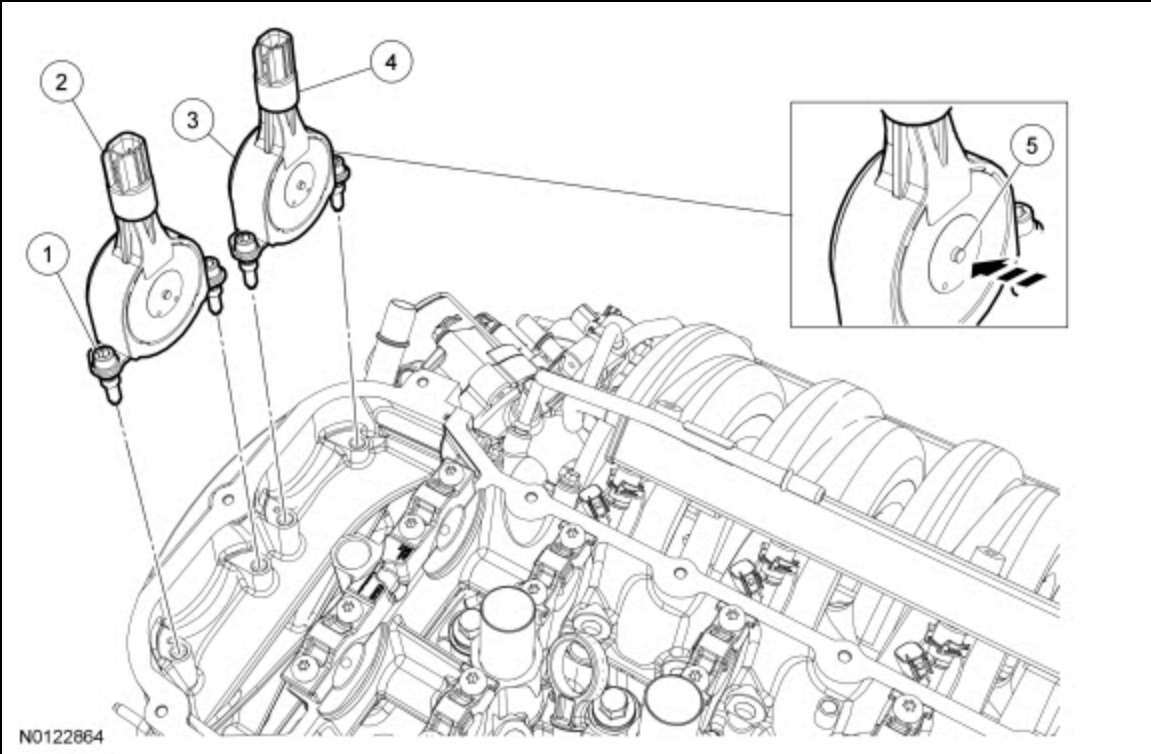
NOTE: The LH or RH valve cover must be removed to perform this procedure. For additional information, refer to Section 303-01C.

NOTE: The bolts are part of the VCT variable force solenoid and cannot be removed.

NOTE: Position the VCT variable force solenoid and tighten the bolts in 2 stages.

- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 30 degrees.

NOTE: LH shown, RH similar.



Item	Part Number	Description
1	W700005 W700005	LH exhaust Variable Camshaft Timing (VCT) variable force solenoid bolt (part of 6M280) (2 required)
2	6M280 6M280	LH exhaust VCT variable force solenoid
3	W700005 W700005	LH intake VCT variable force solenoid bolt (part of 6M280) (2 required)
4	6M280 6M280	LH intake VCT variable force solenoid
5	—	VCT variable force solenoid pin (part of 6M280) (depress prior to installing solenoid)

1. Refer to the procedures and/or exploded views in this section for any Warnings, Notices, Notes, Materials, Specifications, and Special Tools. Items in the exploded views may not be listed in order of removal.